Research Article

Organic honey supplementation reverses pesticide-induced genotoxicity by modulating DNA damage response

Renata Alleva[1], Nicola Manzella[2], Simona Gaetani[2], Veronica Ciarapica[2], Massimo Bracci[2], Maria Fiorenza Caboni[3], Federica Pasini[3], Federica Monaco[2], Monica Amati[2], Battista Borghi[1] and Marco Tomasetti[2]

[1] Department of Anesthesiology Research Unit, IRCCS Orthopaedic Institute Rizzoli, Bologna, Italy
[2] Department of Clinical and Molecular Sciences, Polytechnic University of Marche, Ancona, Italy
[3] Interdepartmental Centre of Agri-food Industrial Research, University of Bologna, Italy

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Scope: Glyphosate (GLY) and organophosphorus insecticides such as chlorpyrifos (CPF) may cause DNA damage and cancer in exposed individuals through mitochondrial dysfunction. Polyphenols ubiquitously present in fruits and vegetables, have been viewed as antioxidant molecules, but also influence mitochondrial homeostasis. Here, honey containing polyphenol compounds was evaluated for its potential protective effect on pesticide-induced genotoxicity.
Methods and results: Honey extracts from four floral organic sources were evaluated for their polyphenol content, antioxidant activity, and potential protective effects on pesticide-related mitochondrial destabilization, reactive oxygen and nitrogen species formation, and DNA damage response in human bronchial epithelial and neuronal cells. The protective effect of honey was, then evaluated in a residential population chronically exposed to pesticides. The four honey types showed a different polyphenol profile associated with a different antioxidant power. The pesticide-induced mitochondrial dysfunction parallels ROS formation from mitochondria (mtROS) and consequent DNA damage. Honey extracts efficiently inhibited pesticide-induced mtROS formation, and reduced DNA damage by upregulation of DNA repair through NFR2. Honey supplementation enhanced DNA repair activity in a residential population chronically exposed to pesticides, which resulted in a marked reduction of pesticide-induced DNA lesions.
Conclusion: These results provide new insight regarding the effect of honey containing polyphenols on pesticide-induced DNA damage response.

Keywords:
DNA damage response / Mitochondria / Organic honey / Pesticides / Polyphenols

Additional supporting information may be found in the online version of this article at the publisher's web-site

Correspondence: Renata Alleva
E-mail: rena.alleva@gmail.com

Abbreviations: **ANOVA**, one-way analysis of variance; **BER**, base excision repair; **CE**, cytoplasm extract; **CPF**, chlorpyrifos; **DCFDA**, dichlorofluorescein diacetate; **DDR**, DNA damage response; **ENDO III**, endonuclease III; **ETC**, electron transport chain; **FPG**, formamido pyrimidine glycosylase; **GLY**, glyphosate; **mtROS**, mitochondrial ROS; **NAC**, N-acethyl cysteine; **NRF2**, nuclear factor erythroid 2-related factor 2; **ORAC**, oxygen radical absorbance capacity; **OGG1**, 8-oxoguanine DNA glycosylase 1; **ROSN**, reactive oxygen and nitrogen species; **SSB**, single strand break; **TCP**, 3,5,6-trichlor-2-piridinol

1 Introduction

Human health is determined by various interacting factors, among which lifestyle, environment, and diet play an important synergistic role in the prevention of several diseases, including cancers. Human intervention trials have provided evidence for protective effects of various polyphenol-rich foods against chronic disease, including cardiovascular disease, neurodegeneration, and cancer [1–4]. While there are considerable data suggesting the benefits of polyphenol rich-diet, conclusions regarding their preventive potential remain

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unresolved due to several limitations in existing studies. Several polyphenols have been shown to effectively modulate pathways that define mitochondrial biogenesis, mitochondrial membrane potential (i.e. mitochondrial permeability transition pore opening and uncoupling effects), mitochondrial electron transport chain (ETC) and ATP synthesis [5]. Polyphenols can act as antioxidants either through a ROS-scavenging mechanism or actions promoted at the ROS-removing level by directly inhibiting the major ROS-forming enzymes. Recent evidence indicates that, by acting at the ETC-complexes level, certain polyphenols would be able to modulate the rate of mitochondrial superoxide production [6]. The pathogenesis of both neurological disorders and cancer has been related to mitochondrial dysfunction [7, 8]. Mitochondrial dysfunction and consequent energy depletion are the major causes of oxidative stress resulting in alteration of ionic homeostasis and causing loss of cellular integrity.

Among various environmental toxicants, pesticides have been shown to affect mitochondria. Exposure to chlorpyrifos (CPF), an insecticide widely used in agriculture, was found to decrease the activity level of Krebs cycle enzymes and ETC protein complexes [9,10]. As well, glyphosate (GLY), the active ingredient in Roundup®, has also been shown to severely deplete Manganese (Mn) levels, associated with Mn-superoxide dismutase suppression, and mitochondrial dysfunction [11].

In this context, honey is a good source of physiologically active natural compounds. Here, honey extracts from four floral organic sources (acacia, chestnut, orange tree, woodland) were evaluated for their polyphenol content, antioxidant activity, and potential protective effects on pesticide-related mitochondrial destabilization, reactive oxygen and nitrogen species (ROSN) formation, and DNA damage response (DDR) in human bronchial epithelial cells (pesticide penetration way) and neuronal cells (pesticide target cells). The protective effect of honey was, also in vivo evaluated in a population chronically exposed to pesticides.

2 Materials and methods

2.1 Sample extraction

Phenolic compounds for HPLC analysis were extracted from honey (acacia, Ext-1; chestnut, Ext-2; orange tree, Ext-3; woodland, Ext-4) as described previously [12]. About 3 g of honey was dissolved in 15 mL of acidified water (pH = 2 with HCl), stirring at room temperature until completely fluid. The solution was mixed with 4 g Amberlite XAD-2 (pore size 9 nm, particle size 0.3–1.2 mm, Sigma) and stirred for 10 min. The Amberlite particles were then packed in a glass column (30 cm × 3 cm), washed with acidified water (pH = 2 with HCl, 10 mL) and subsequently rinsed with distilled water (30 mL). The whole phenolic fraction was then eluted with methanol (3 mL) and taken to dryness by vacuum Concentrators (SpeedVac, Thermo Scientific). The residue was resuspended in distilled water to a concentration of 100 mg/mL.

2.2 HPLC–DAD–MS analysis

All HPLC analyses were performed using an HP 1100 Series Instrument (Hewlett–Packard, Wilmington, DE) equipped with a binary pump (model G1312A) delivery system, a degasser (model G1322A), an autosampler (Automatic Liquid Sampler, ALS, model G1313A), a HP diode-array UV–VIS detector (DAD, model G1315A), and a HP-single-quadrupole mass spectrometer detector (MS, model G1946A). Separations were carried out on a reverse phase column Eclipse XDB-C18, 5 μm 250, 3.0 mm id, (Agilent Technologies, Santa Clara, CA, USA) with a Securityguard precolumn filter as previously described [12]. The quantitative analyses were carried out in triplicate and expressed as micrograms per 100 g of honey.

2.3 Oxygen radical absorbance capacity assay

The oxygen radical absorbance capacity (ORAC) assay was performed by OxiSelect™ ORAC Activity Assay Kit (Cell Biolabs) according to the manufacturer's instructions. Fluorescence was read (λ_{exc} 485 nm and λ_{em} 520 nm) every 20 min for a total of 160 min using a fluorescence plate reader (Infinite F200 PRO, Tecan). The area under the curve was calculated for each sample and compared with that of Trolox. ORAC values are expressed as mmol Trolox equivalent.

2.4 Cell culture and treatment

Human bronchial epithelial cells (BEAS-2B) and neuronal SHSY-5Y cell line were used as a model of in vitro evaluation of mutagenesis and neurodegenerative disease studies, respectively. BEAS-2B (ATCC® CRL9609™), and SHSY-5Y cells (ATCC® CRL2266™) were grown in the DMEM medium with antibiotics and 10% FBS, and regularly checked for absence of mycoplasma contamination using the PCR Mycoplasma Test. Cells were treated with pesticides (ranging 10–1000 μM) in absence or presence of honey-extracts (5 μg/mL) added 2 h before pesticide treatments. Chronic exposure to pesticides was established by incubating BEAS-2B and SHSY-5Y cells with GLY and CPF (100 μM in fresh medium, three times a week) over a period of 6 months.

2.5 Cell proliferation assay and mitochondrial activity

Cells were seeded at 10^4 cells/well in a 96-well plate, allowed to attach overnight, and over time (24-48-72 h) treated with increasing concentrations of pesticides (0, 10, 100, 500, 1000 μM). After treatment, 50 μL of crystal violet (2% crystal violet in 2% ethanol) were added, and incubated for 5 min. After four washes, 200 μL of isopropanol was added to dissolve the crystals, and absorbance read at 570 nm in an ELISA plate reader (Sunrise, Tecan, Männedorf, Swiss), and the results expressed as rate of proliferation (abs/time).

For mitochondrial activity, the resazurin assay was performed. Cells were incubated with resazurin (6 μM) in the presence and absence of pesticides (0, 10, 100, 500, 1000 μM). Fluorescence intensity was read at 0–240 min in a fluorescence plate reader (Infinite F200 PRO, Sunrise, Tecan, Männedorf, Swiss). The excitation and emission filters were set at 485 and 530 nm, respectively. The results were normalized to the total protein using the Bradford assay (Sigma), and expressed as rate (MFI/min).

2.6 Assessment of ROSN

Intracellular ROSN levels were estimated using the fluorescent dye 2′7′-dichlorofluorescein diacetate (DCFDA; oxidized by hydrogen peroxide to DCF). BEAS-2B and SHSY-5Y cells (2×10^5) were seeded in six-well plates, supplemented with 20 μM DCFDA per well, and treated with pesticides (10 and 100 μM) in presence or absence of honey-extracts (5 μg/mL). After treatment the florescent probe was removed, the cells washed, and resuspended in PBS, and analyzed by flow cytometry (FACS Calibur, Becton Dickinson). The level of ROSN was expressed fold increase in fluorescence respect to control (untreated cells).

2.7 DNA damage

DNA breaks and oxidized purine and pyrimidine bases were measured using the Comet assay described elsewhere [13]. Briefly, lymphocytes or treated and control cells, were embedded in agarose on a microscope slide, lysed with Triton X-100 and 2.5 M NaCl to produce nucleoids and treated with 0.3 M NaOH/1 mM EDTA before electrophoresis in this solution. Oxidized bases were detected by including an extra step, in which nucleoids in the gel are digested with a repair endonuclease specific for oxidized pyrimidines (endonuclease III, ENDO III) or recognizing altered purines, including 8-oxodG (FPG protein). Slides were incubated for 30 min with 50 μL of either buffer, formamido pyrimidine glycosylase (FPG) and ENDO III (generously gift from Prof. Andrew Collins, University of Oslo, Oslo, Norway). DNA single strand breaks (SSBs), with or without enzymatic treatment, were estimated as arbitrary units (au). Oxidized purine and pyrimidine bases were calculated by subtracting the value without enzyme incubation (i.e. SSBs) from the value with enzyme incubation. The extent of DNA migration was evaluated by visual scoring by an independent observer.

2.8 DNA repair assay

The activity of DNA repair was evaluated as capacity of cell extract to repair the oxidized purine (8-oxodG) introducing DNA breaks detected as previously described [13, 14]. Briefly, cell extract was prepared from 2.0×10^6 cells in extract buffer (45 mM Hepes, 0.4 M KCl, 1.0 mM EDTA, 1 mM DTT, and 10% v/v glycerol adjusted to pH 7.8 with KOH), and was cryopreserved at –80°C until used in the DNA repair assay. The protein extract (50 μg) was added over a slide with a substrate consisting of nucleoid DNA with oxidized purine bases from A549 cells previously exposed to Ro 19–8022 (generously gift from Hoffmann-LaRoche, Basel, Switzerland) plus light, which specifically induces 8-oxodG formation. The slides were incubated for 45 min at 37°C and comet assay (alkaline unwinding, electrophoresis, neutralization, staining, and evaluation) was carried out as described above. The time-course of break production on the specifically damaged DNA substrate is a measure of repair capability.

The ability of cells to repair was also evaluated by measuring residual DNA damage over a period of incubation. Briefly, cells were seeded in a 24-well plate, and damage to DNA was introduced by incubating them with 100 μM H_2O_2 in PBS for 5 min on ice (Tb). After treatment, the PBS was replaced with RPMI-1640 medium containing 10% fetal bovine serum and incubated at 37°C for 6 h. At regular interval times (Tn = 30, 60, 180, 360 min), the cells were collected and SSBs, FPG-sites, ENDO III-sites evaluated by comet assay. Residual damage (%) at different time points was determined as follows: 100 × [DNA damage at time T after treatment (Tn)—DNA damage of untreated cells (T0)/DNA damage immediately after treatment (Tb)—DNA damage of untreated cells (T0)]. The $t_{1/2}$ (h) were determined using plots of data obtained from the initial time (t_i) of induced DNA damage level (C_i) to the final time (t_f) when damage was completely repaired (C_f), according to the following formulas: $K = (\log C_i – \log C_f)/(t_f – t_i)$, $t_{1/2} = \log 2/K$.

2.9 Western blot analysis

Cells (3×10^5 per well in six-well plates) were harvested, and the pellet resuspended in the cytoplasm extract (CE) buffer (10 mM HEPES, 60 mM KCl, 1 mM EDTA, 0.075% v/v NP40, 1 mM DTT, and 1 mM PMSF, adjusted to pH 7.6). After 5 min incubation on ice, the pellet was centrifuged at 1500 rpms for 5 min, and the supernatant containing the cytosolic fraction obtained. The remaining pellet was washed in CE buffer, and lysed in the nuclear extract buffer (20 mM Tris-HCl, 420 mM NaCl, 1.5 mM MgCl2, 0.2 mM EDTA, 1 mM PMSF, and 25% v/v glycerol, adjusted to pH 8.0). After 20 min incubation on ice, the pellet was centrifuged at 12 000 rpms for 10 min at 4°C, and the supernatant was collected. The nuclear extracts were stored at –80°C until used. For Western blot analysis, protein samples (50 μg per lane) were resolved using

4–12% SDS-PAGE (Life Technologies), and transferred to nitrocellulose membranes, and incubated overnight with anti-NRF2 (Cell Signaling Technology, Danvers, MA) and anti-OGG1 (Origene, Rockville, MD). Lamin (Bethyl, Montgomery, TX, USA) was used as loading control. After incubation with an HRP-conjugated secondary IgG (Sigma), the blots were developed using the ECL detection system (Pierce Biotechnology, Rockford, IL, USA). Band intensities were visualized by ChemiDoc using the Quantity One software (BioRad, Hercules, CA).

2.10 Ethics statement

All subjects filled a questionnaire including their informed consent. The study was carried out according to the Helsinki Declaration and the samples were processed under approval of the written consent statement of the Ethical Regional Committee, Italy (n° 211584), according to Ministerial Decree (DM 12/05/2006).

2.11 Recruitment of study participants

Residents, belonging to health committee, who presented themselves for a routine checkup were recruited at the Medical Centre of Tuenno (TR), Italy. The enrolled subjects were residents in the Val di Non (Trento, Italy), an area of intensive culture of apple orchards. In this area, from March to middle of October apple orchards undergo to a series of pesticide treatments as spray solution preparation, which includes fungicides (boscalid, captan, fluazinam, iprodione, penconazole), herbicides (glyphosate), acaridae (Bromopropylate), and insecticides (chlorpirifos etile, piretrine, metossifenozide). Therefore, more than 30 treatments, often as mix of different substances, are used within the year, with major exposure at the May–June period. Due to the conformation of the area, which is constituted by valley closed to mountains, the pesticides remain in the environment affecting also the residential population. Residents chronically exposed to pesticides were 34 subjects, who consented to participate in the study, and included teachers and employees, engineers, architects, all subjected to passive exposure to pesticides. The participants were interviewed by trained personnel and answered a detailed questionnaire that included life-style including dietary consumption, smoking, environmental, and occupational exposure.

The control group consisted of healthy subjects ($n = 40$), having no previous or current occupational/environmental exposure to pesticides. The subjects were undergoing medical check-up at the Clinic of Occupational Medicine of the University Hospital of Ancona, Italy. None of them had ever been exposed to pesticides as documented by their environmental and occupational histories.

2.12 Sample collection

Three periods of exposure to pesticides were surveyed: no pesticides-exposure, when the pesticides were not used (P0, February–March period), low-exposure, when pesticides were used occasionally (P1, October–November period), and high exposure, in which pesticides were used regularly on a weekly basis (P2, May–June period). Blood samples were collected in fasting subjects at P0 and at the P1 and P2 periods, before (S0) and after two weeks of organic honey dietary intake (50 g/day, S1) administered during the day at the morning and afternoon times in substitution of other sweeteners. Based on the polyphenol content and antioxidant capacity, woodland honey was used for subject supplementation. The subjects were asked to follow a dietary plan at controlled intake of polyphenols (800 $\pm$ 20 mg/day, using the phenol-explorer database on polyphenol content in food) a week before sample collection and during the period of honey supplementation. All participants provided for blood sample for each period at the time of interview. Blood samples were collected into EDTA tubes and used for lymphocyte isolation as previously described [13, 14]. Morning urine samples (~50 mL) were collected in polypropylene containers (Starplex Scientific, Canada) at all three exposure periods (P0, P1, and P2), before and after honey supplementation (S0, S1) and stored frozen at –80°C until analysis. The control subjects provided for urine samples at the moment of enrollment.

2.13 Air sample collection

Air sampling was conducted for 24 h (9.00 AM to 9.00 AM of the day after) in three areas at the high-exposure period (P2). Air samples were collected near the apple orchards by using sampling pumps connected to the sampler (Solid sorbent tube; OVS-2 tube: 13 mm quartz; XAD-2, 270 mg/140 mg). The pump was set at a flow rate of 2 L/min and the sampler was set to connect along the calibrated sampling pump with tube. Pump flow-calibration checks were performed before and at the end of the sampling period. At the completion of sampling, the sampler was capped at both ends with plastic caps and packed for shipment. After collection, the samples were labeled and frozen in an icebox to be sent to the laboratory. Samplers were used and the solvent was extracted and analyzed by GC-ICP-MS (Agilent 7000C GC Triple Quadrupole GC/MS system).

2.14 3,5,6-trichlor-2-piridinol analysis

Biological monitoring of exposure was performed by determining chlorpirifos ethyl metabolite (3,5,6-trichlor-2-piridinol, TCP) in urine samples according to the method previously described [15]. The results were expressed as a function of creatinine (creat) levels (mg/g creat).

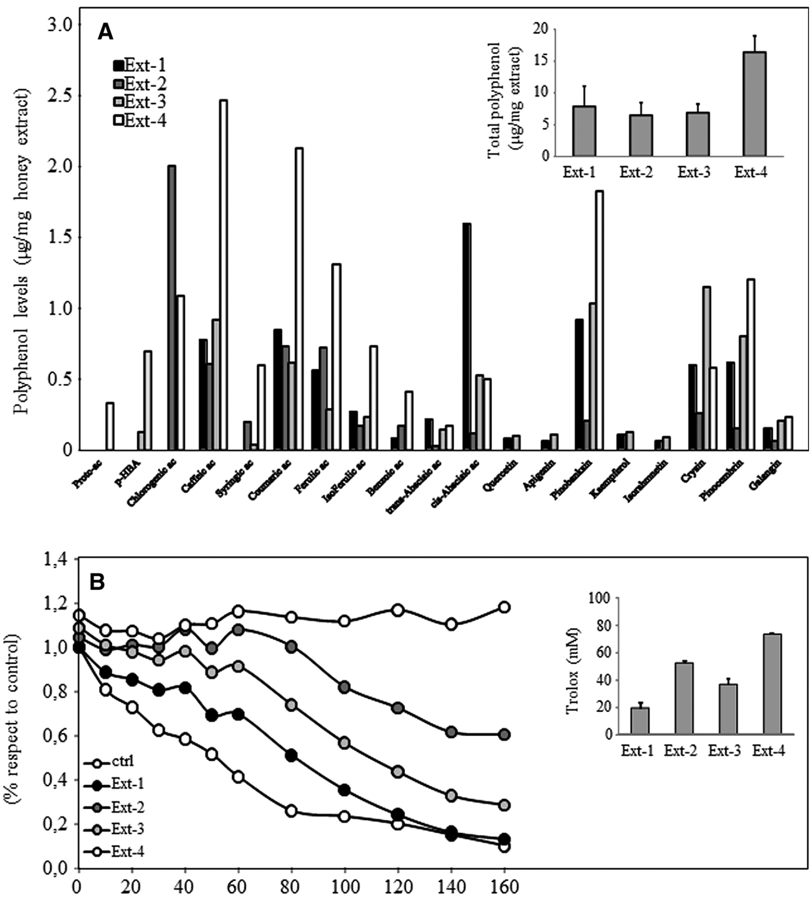

Figure 1. Polyphenol profile of honey extracts, and their antioxidant capacity. (A) Extracts from honey (acacia, Ext-1; chestnut, Ext-2; orange tree, Ext-3; woodland, Ext-4) were evaluated for their polyphenol types and content. The total polyphenol level for each extracts is shown in the insert. (B) The antioxidant property of honey extracts was evaluated by oxygen radical absorbance capacity (ORAC) assay, and the total antioxidant activity expressed as mM trolox (insert). The results are the mean ± SD of three experiments.

2.15 Total polyphenol detection in urines

Total polyphenols were detected in urine samples collected at the periods of pesticide exposure, before and after honey supplementation. Urine samples (1 mL) were centrifuged and acidified with 34 μL of hydrochloric acid at 35%; they were used to load to a Sep-Pack C 18 cartridge (Waters, USA), which was preequilibrated by methanol (3 mL) followed by acidified water (3 mL, pH = 2). The phenolic content was eluted with 80% v/v methanol (1 mL). The total polyphenols were detected by a Folin-Ciocalteu method previously described [16], and expressed as mg gallic acid equivalent per creatinine levels (mg/g creat).

2.16 Statistical analysis

Results were expressed as mean ± SD, or median (25° percentile–75° percentile). Comparisons among groups of data were made using one-way analysis of variance (ANOVA) with Tukey post-hoc analysis. The two-tailed Student's *t*-test was used to compare two groups. ANOVA repeated measure with Sidak post-hoc test was used to evaluate differences among the P0, P1-S0, P1-S1, P2-S0, and P2-S1 time points. Multiple regression analysis was used to model DDR parameters, such as FPG, ENDO III, and DNA repair activity as function of honey supplementation taking into account age, gender, body mass index (BMI), and smoking. The data were analyzed by the Statistical Package Social Sciences (version 19) software (SPSS, Chicago, IL, USA) and *p*-values less than 0.05 were considered significant.

3 Results

3.1 Polyphenol profile and antioxidant capacity of honey extracts

All the tested organic honey samples had similar but quantitatively different phenolic profiles and up to 20 peaks could be assigned to phenolic compounds and identified as phenolic acids. As reported in Fig. 1A, the Ext-4 was richer in polyphenols with respect to other extracts (Ext-1, Ext-2, and Ext-3), showing high levels of caffeic acid, coumaric acid, Ferulic acid, iso-Ferulic acid, pinobanksin, and pinocembrin. The higher content of polyphenols reflected a higher antioxidant capacity followed by Ext-4, Ext-2, Ext-3, and Ext-1 (Fig. 1B).

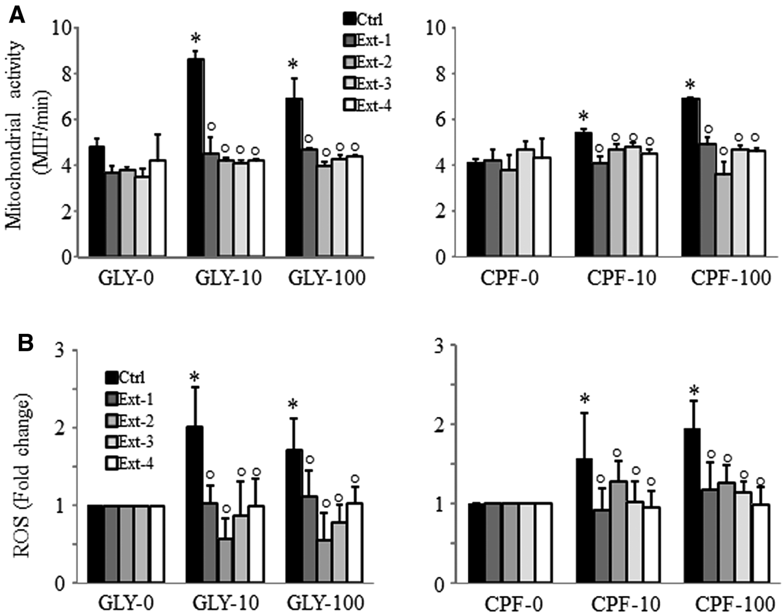

Figure 2. Effects of honey-extracts on pesticides-induced mitochondrial destabilization and ROS formation. BEAS-2B cells were 2 h pretreated with honey extracts (5 μg/mL) followed by 10 and 100 μM glyphosate (GLY, left-panel) or 10 and 100 μM chlorpyrifos (CPF, right-panel). Control cells received GLY or CPF treatment (10 and 100 μM). After 24 h of incubation, mitochondrial activity (A) and ROSN formation (B) were evaluated. The results are the mean ± SD of three experiments performed in duplicate. Comparisons among groups were determined by one-way ANOVA with Tukey post hoc analysis. The symbol "*" indicates significant differences compared with untreated cells (GLY-0, CPF-0), the symbol "°" significance compared with control (Ctrl), with $p < 0.05$.

3.2 Honey extracts attenuate mitochondrial destabilization induced by pesticides

BEAS-2B and SHSY-5Y cell lines were treated with increased concentration of GLY and CPF ranging 10–1000 μM, and cell proliferation, mitochondrial activity evaluated. GLY induced cell proliferation in a dose-dependent manner, associated with increased mitochondrial activity at low concentration, which was reduced at increased doses (>100 μM) in BEAS-2B cells (Supporting Information Fig. S1A). No effect of GLY was found in SHSY-5Y cells (Supporting Information Fig. 1B). Conversely, CPF inhibited cell proliferation in both cell lines, while inducing mitochondrial activity at concentrations below 100 μM. Therefore, two subtoxic doses of GLY and CPF, 10 μM (low-dose) and 100 μM (high-dose) were used to establish a genotoxic stimulus model. As reported in Supporting Information Fig. 1C, both pesticides at low-dose, and in a major extent at high-dose induced ROSN formation measured in terms of fluorescence by DCF. To investigate the effect of honey extracts, BEAS-2B and SHSY-5Ycells were treated with GLY and CPF (10 and 100 μM) over time in presence or absence of extracts (5 μg/mL) and mitochondrial activity and ROSN formation evaluated. The increased mitochondrial activity and ROSN formation induced by pesticides (GLY and CPF), peaking at 1–3 h and 24 h of pesticide exposure (Supporting Information Fig. 2A, B), were reversed by honey-extracts in both cell lines (Fig. 2A, B, and Supporting Information Fig. 3A, B).

3.3 Honey extracts inhibit pesticides-induced DNA damage

Mitochondria are the major producer of intracellular ROS, and have been linked to the cause of aging and other chronic diseases [17]. To determine a direct relationship between intracellular ROS levels and pesticides-induced mitochondrial destabilization, the cells were pretreated with the superoxide scavenger Tiron (10 mM), ROS scavenger N-acethyl cysteine (NAC, 10 mM), honey-extract (5 μg/mL), and ROSN formation was induced by GLY and CPF treatments (10 and 100 μM) for 24 h. As reported in Fig. 3A, the ROSN induced by pesticides were inhibited by Tiron, and in a major extent by NAC and honey-extract. The inhibition of ROSN by honey-extract paralleled the reduction of pesticides-induced DNA damage evaluated as single strand breaks (SSBs) and oxidized DNA base accumulation (FPG-sites and ENDOIII-sites) both in BEAS-2B cells (Fig. 3B, C) and SHSY-5Y cells (Fig. 3D). Thus, suggesting that pesticides affect mitochondria homeostasis by inducing ROS production (mtROS), which in turn induce DNA damage. This process was significantly inhibited by honey extracts.

3.4 Honey extracts rescue cells from pesticides-induced DNA repair inhibition

Endogenous DNA lesions are the results of a balance between DNA damage and DNA repair (steady state). The failure to detect and accurately repair these lesions can give rise to cells with high levels of endogenous DNA damage, deleterious mutations, or genomic aberrations. Such genomic instability can lead to the activation of specific signaling pathways, including the DDR. Incubation of cells with honey-extracts (5 μg/mL) enhanced DNA repair activity in BEAS-2B and slightly in SHSY-5Y cells (Fig. 4A). As reported in Fig. 4B, the cell showing higher DNA repair activity with respect to their nonsupplemented counterparts resulted in an increased capacity to repair DNA damage induced by hydrogen

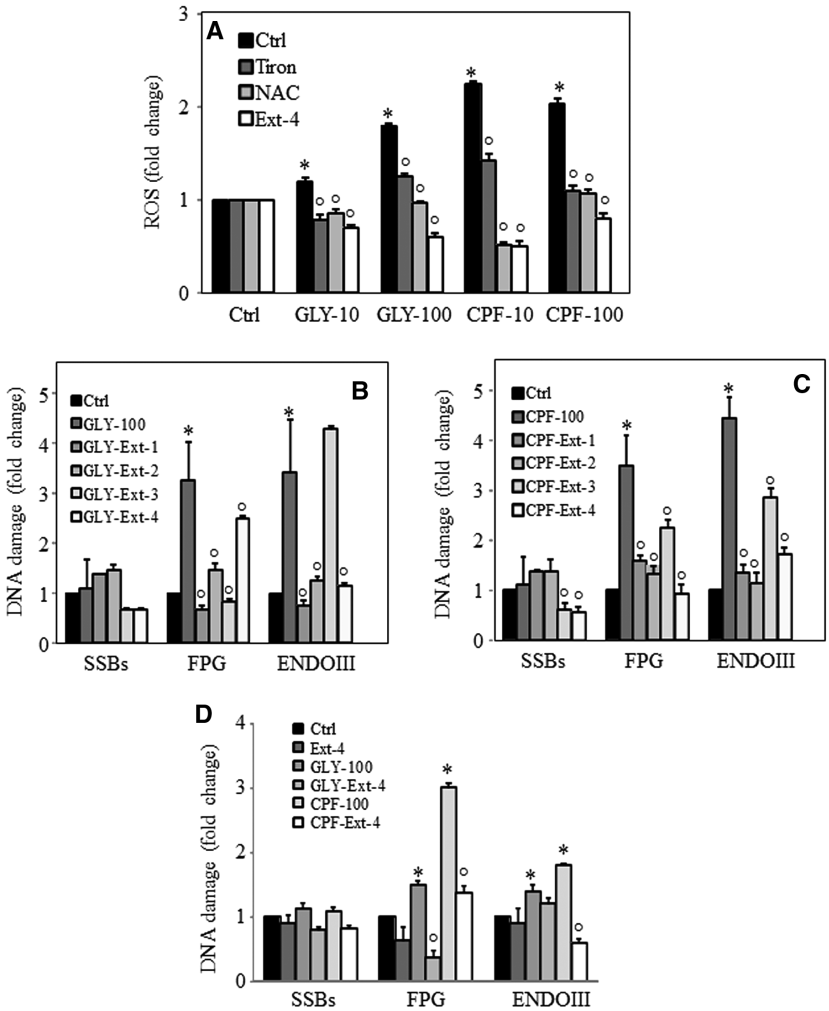

Figure 3. Effects on ROS inhibitors Tiron and NAC, and honey-extract to pesticides-induced ROS formation. (A) BEAS-2B cells were treated with 10–100 μM glyphosate (GLY) or 10–100 μM chlorpyrifos (CPF) in the presence or absence of Tiron (10 mM), NAC (10 mM), and honey extract (5 μg/mL). (B) BEAS-2B cells were treated with 10–100 μM GLY (left-panel) or 10–100 μM CPF (right-panel) in the presence or absence of honey-extracts (5 μg/mL) and DNA damage evaluated as single strand breaks (SSBs), FPG, and ENDOIII lesions. (C) SHSY-5Y cells were treated with GLY (100 μM) or CPF (100 μM) in presence or absence of honey-extract (5 μg/mL) and DNA damage evaluated. The results are expressed as fold change of control of three independent experiments. Comparisons among groups were determined by one-way ANOVA with Tukey post hoc analysis. The symbol "*" indicates significant differences compared with untreated cells (Ctrl), the symbol "°" significance between pesticide-treated cells and cell incubated with ROS inhibitors (Tiron, NAC) or honey extract, with $p < 0.05$.

peroxide (H_2O_2). Conversely, both GLY and CPF incubation (100 μM for 24 h) inhibited cellular DNA repair activity, which was reversed by honey-extract (Fig. 4C). In addition, chronic exposure to pesticides was established by incubating BEAS-2B and SHSY-5Y cells with GLY and CPF (100 μM in fresh medium, three times a week) over a period of 6 months. Cells chronically exposed to pesticides showed lower DNA repair activity with respect to untreated cells (Controls). The incubation of honey-extract increased DNA repair activity at levels similar to control cells (Fig. 4D).

3.5 NRF2 plays an important role in the modulation of the DNA repair activity

Nuclear factor erythroid 2-related factor 2 (NRF2) is a known regulator of the antioxidant response [18]. NRF2-mediated regulation of protective enzymes like oxidative DNA damage repair enzyme 8-oxoguanine DNA glycosylase 1 (OGG1) [19]. OGG1 plays an essential role in the removal of 8-oxoguanines (FPG-sites) from nuclei. We examined the nuclear translocation of NRF2 and OGG1 in BEAS-2B and SHSY-5Y cells treated with GLY and CPF (100 μM, 24 h) with and without honey extract (5 μg/mL), and compared it with NRF2-OGG1 expression in their control cells. Pesticide incubation induces nuclear translocation of NRF2, which paralleled increased OGG1 nuclear expression (Fig. 5A, B). Honey extract from woodland (Ext-4) alone or in combination with pesticides significantly increased nuclear protein expression of NRF2 and OGG1 (Fig. 5A, B). Notably, the SHSY-5Y cells were less reactive to pesticide- and honey extract-induced NRF2/OGG1 nuclear translocation.

3.6 Honey-supplementation enhances DDR in residents chronically exposed to pesticides

The effect of honey containing polyphenols on DDR was in vivo evaluated in a population chronically exposed to pesticides. Residents from an area at intensive use of pesticides (Val di Non, Trento, Italy), and subjects living in a pesticide-free area, negative for urine TCP, were included in the study.

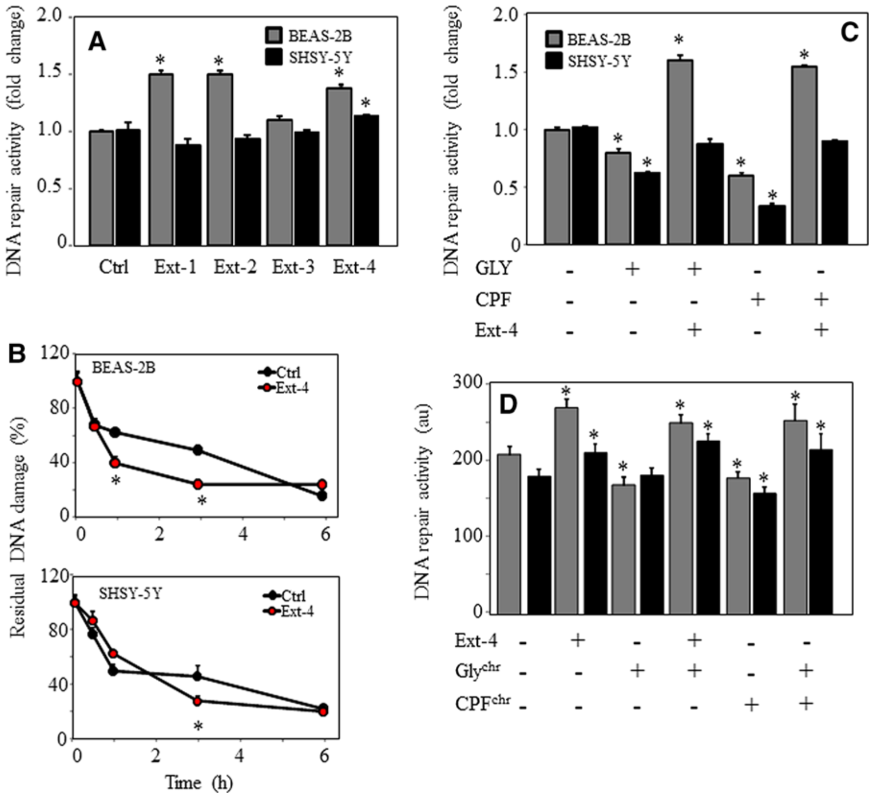

Figure 4. Effect of honey-extracts on DNA repair activity. Human bronchial epithelial cells (BEAS-2B) and neuronal SHSY-5Y cell line were overnight incubated with honey extracts and the DNA repair activity was evaluated in cellular extracts (A) or as kinetics of DNA strand break rejoining in cells after exposure to hydrogen peroxide (H_2O_2, 100 μM). (B) The kinetics of DNA repair was calculated as a percentage of residue damage with respect to the time of maximum damage (Tn). Residual damage (%) = 100 × [DNA damage at time T after treatment (Tn)–-DNA damage of untreated cells (T0)/DNA damage immediately after treatment (Tb)—DNA damage of untreated cells (T0)]. BEAS-2B and SHSY-5Y cells were treated with 10–100 μM glyphosate (GLY) or 10–100 μM chlorpyrifos (CPF) for 24 h (C) or chronically exposed to both pesticides (D) in the presence or absence of honey extract (5 μg/mL, 24 h), and DNA repair activity evaluated. The symbol "*" indicates significant differences compared with control (Ctrl) and treated cells, with $p < 0.05$.

The residents were active, mainly consumed fish and organic foods, with low consumption of meat and sweets (Supporting Information Fig. 4). The demographic and anthropometric characteristics are shown in Table 1A. Based on the calendar of treatments the DDR was evaluated in the population at three periods: P0, no pesticides-exposure; P1, low-pesticide exposure, and P2, high-pesticide exposure. At the P1 and P2 periods, the DDR was evaluated in subjects before and after 2-week honey supplementation (50 gr/Die). The polyphenol content was evaluated as total polyphenols in the urine samples during the periods of pesticide exposure, before and after honey supplementation (Table 1B). All the residents were positive for the urine metabolite TCP in all periods analyzed, showing high TCP levels at high-exposure period (Table 1C).

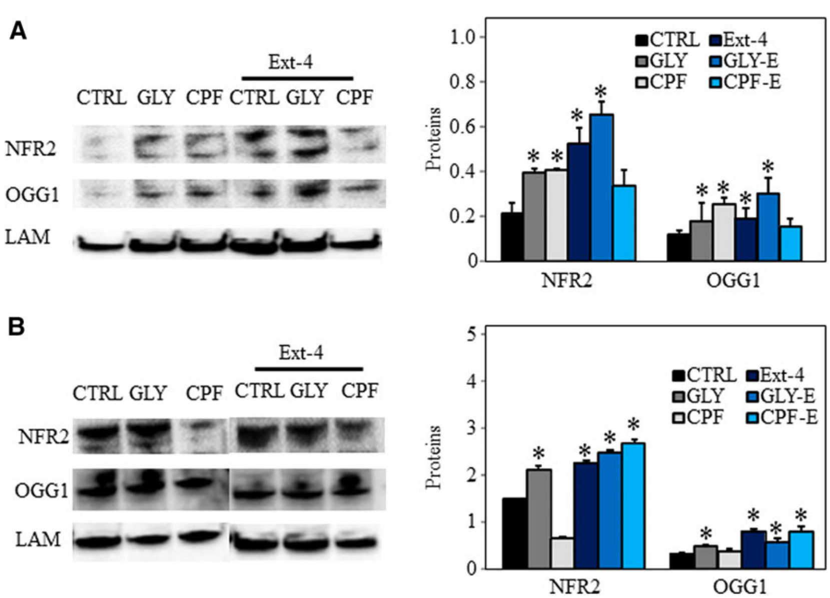

Figure 5. Cellular NFR2 activation and OGG1 expression are induced in response to pesticides and honey extract treatment. Human bronchial epithelial cells (BEAS-2B) and neuronal SHSY-5Y cell line were treated with glyphosate (GLY, 100 μM) and chlorpyrifos (CPF, 100 μM) with or without honey extract (Ext-4) preincubation (2 h). The NFR2 and OGG1 nuclear translocation were evaluated in BEAS-2B cells (A) and SHSY-5Y cells (B) by Western blot (lest-panel), and the levels of the bands quantified by densitometry analysis (right-panel). Comparisons among groups were determined by one-way ANOVA with Tukey post hoc analysis. The symbol "*" indicates significant differences compared with untreated cells (Ctrl).

Table 1. Demographic and anthropometric characteristics of the recruited subjects, and biological/environmental exposure to pesticides

A

Demographic/anthropometric indices	Pesticides exposed population ($n = 34$)			Pesticides nonexposed population ($n = 40$)		
	Median	25°–75°	%	Median	25°–75°	%
Age (years)	46	15–55		35	30–55	
Sex (male/female)			41/59			48/52
Weight (kg)	58	50–70		56	55–72	
Height (cm)	166	151–174		159	162–175	
BMI (Kg/m^2)	22	20–23		21	20–22	
Smoking (yes/no)			22/78			12/82

B

Periods of exposure to pesticides

	P0	P1		P2	
		S0	S1	S0	S1
	Median (25°–75°)	Median (25°–75°)	Median (25°–75°)	Median (25°–75°)	Median (25°–75°)
Tot Polyph. (mg/g creat)	95 (75–142)	110 (76–142)	**229 (99–268)°**	119 (87–162)	**268 (211–300)°**

C

Periods of exposure to pesticides

	P0 Median (25°–75°)	P1 Median (25°–75°)	P2 Median (25°–75°)
Biological pesticide level			
TCP (μg/g creat)	2.1 (1.9–3.5)	3.2 (2.7–3.6)*	4.4 (3.3–5.0)*
Environmental pesticides level (ng/m^3)			
Chlorpyrifos ethyl	—	—	5.1 (0.7–7.4)
Difenoconazol	—	—	2.2 (0.7–2.8)
Fluazinam	—	—	18.3 (17.6–19.4)
Methoxyfenozide	—	—	0.5 (0.3–0.8)

TCP, 3,5,6-trichlor-2-piridinol; creat, creatinine.
P0, no-exposure period; P1, low-exposure period; P2, high-exposure period.
S0, before honey supplementation; S1, after honey supplementation.
Data are shown as median [25° percentile-75° percentile] for continuous variables and percentages for categorical variables. *P1 and P2 versus P0; °S0 versus S1, $p < 0.05$.

Among the multiresidual pesticides analyzed (see list of compounds in Supporting Information Fig. 5), only chlorpyrifos ethyl, Difenoconazole, Fluazinam, and methoxyfenozide have been detected in air samples (Table 1C). The DDR parameters (FPG, ENDO III, and DNA repair activity) evaluated in function of honey supplementation were independent from Age, gender, BMI, and smoking as evaluated by multiple regression analysis (data not shown). In the period of no-exposure, the population chronically exposed to pesticides showed reduced DNA repair activity associated with a defective DNA damage recovery with respect to control subjects (Fig. 6). Although DNA repair was low in these subjects, they showed comparable levels of FPG and ENDOIII lesions with respect to controls (Fig. 6). The pesticide exposure further reduced the DNA repair activity, which was associated with increased FPG and ENDOIII sites formation. Notably, the honey supplementation enhanced DNA repair activity, thus resulting in a marked reduction of FPG and ENDOIII lesion accumulation.

4 Discussion

Honey is a polyphenol-rich food, which has been shown to exhibit several biological activities, including antimicrobial [20], anti-inflammatory [21], antioxidant [22], cardio-protective properties [23], and anticancer [24]. Compelling evidence has shown that the polyphenol-enriched fraction from honey can suppress cancer in an in vivo model [25, 26]. To date, approximately 300 varieties of honey have been identified, which are characterized by a different mixture of approximately 30 different polyphenols [27, 28]. The chemical composition

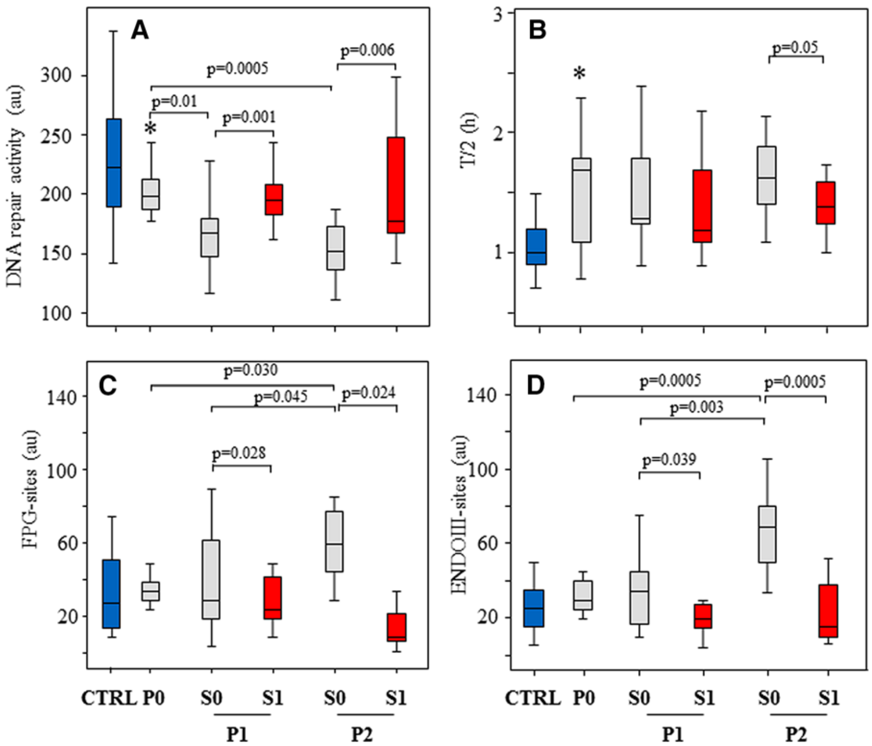

Figure 6. Effect of honey supplementation on the DNA damage response (DDR) in residents chronically exposed to pesticides. A population chronically exposed to pesticides was enrolled at three different periods of pesticide exposure: no pesticides-exposure, (P0, February–March), low-exposure, (P1, October–November), and high exposure, (P2, May–June). In the P1 and P2 periods, DNA repair activity (A), time to repair 50% of H_2O_2-induced DNA damage ($t_{1/2}$) (B), FPG-sites (C), and ENDOIII-sites (D) were evaluated in pesticide-exposed residents before (S0) and after two weeks honey supplementation (50 g/day) (S1). Pesticide nonexposed subjects were used as controls (CTRL). Comparisons among groups were evaluated by ANOVA repeated measure with Sidak post-hoc test, with $p < 0.05$.

contributes to honey bioactivity. In the present study, we evaluated the extracts from four organic honey varieties (acacia, chestnut, orange tree, woodland) for their polyphenol composition, antioxidant effects and effect on toxicity induced by pesticides. Even though the four honey extracts had different antioxidant activity, reflecting the different polyphenol content and profile (*cf* Fig. 1), they showed comparable effect on cell genotoxicity induced by pesticide exposure.

Pesticides are compounds highly used in conventional agriculture practice, and multiresidue of pesticides was found in nonorganic fruit, vegetable, and honeys [29, 30]. Environmental, occupational, and food exposure have been associated with health risk [31–34]. During occupational and residential exposure, pesticides are prevalently absorbed through the airways, and the neurons are one target of these compounds. Prospective cohort studies have shown that prenatal or early childhood exposure to CPF causes adverse neurodevelopmental outcomes, such as decreased head circumference, poorer intellectual development, and cognitive deficits, mental and motor delays, as well as neurodegenerative diseases [35, 36]. Meta-analyses showed that phenoxy herbicides, carbamate insecticides, organophosphorus insecticides and the active ingredient lindane, an organochlorine insecticide, were positively associated with non-Hodgkin lymphoma; B-cell lymphoma was positively associated with phenoxy herbicides and the herbicide glyphosate [37]. Here, bronchial BEAS-2B cell (airway cells) and neuronal SHSY-5Y cell line (target cells) were used as a model to evaluate in vitro pesticide exposure. The two cell lines differently responded to pesticides (GLY and CPF). GLY induced cell proliferation in BEAS-2B cells, even at low dose, but not in SHSY-5Y cells. Conversely, CPF decreased cell proliferation in a dose-dependent manner in both cell lines. Mitochondrial damage has been implicated to play a key role in pesticide-induced neurotoxicity [38, 39]. We showed that GLY and CPF treatment caused increased mitochondrial activity at low doses, while inhibiting mitochondrial activity at higher concentrations. The altered mitochondrial dysfunction was associated with ROSN generation in both cell lines (*cf* Supporting Information Fig. 1). In particular, superoxide production by the mitochondrial respiratory chain can increase dramatically when the mitochondrial membrane potential is high (high mitochondrial activity) associated with low ATP production [40]. Pesticides mainly induced ROS through mitochondrial destabilization, most of ROS where reduced by the superoxide scavenger Tiron. The mtROS produced paralleled DNA damage in both cell lines (cf. Fig. 2). The honey extracts markedly restored the mitochondrial activity, and inhibited ROS-induced DNA damage (cf. Figs. 2 and 3).

Genome instability is a prerequisite for the development of degenerative and cancer diseases. It occurs when genome maintenance systems fail to safeguard the genome's integrity as consequence of inherited defects, or induced via exposure to environmental agents [41]. Various DNA repair pathways prevent the persistence of such DNA lesions, maintaining genome integrity. Alterations to the epigenome may also lead to genome instability in an indirect way. Epigenetic changes can affect DNA repair efficiency and fidelity by changing the expression of DNA repair genes [42, 43]. In the last decade, a number of studies have shown that nutrients can affect metabolic traits by altering the structure of chromatin and directly regulate both transcription and translational processes [44]. In this context, dietary polyphenol-targeted epigenetics becomes an attractive approach for disease prevention and intervention [45]. Here, we found that honey-polyphenols modulate the activity of DNA repair involved in the DNA base excision repair (BER) system. Honey extracts by themselves increased DNA repair activity, associated with an enhanced kinetic to repair the damaged DNA in BEAS-2B cells. In this context, honey extracts reversed the reduced DNA repair activity induced by pesticide exposure. This phenomenon was not observed in neuronal cells, even though the honey extracts prevented the downregulation of DNA repair activity mediated by pesticides in these cells (cf. Fig. 4).

The DNA glycosylase OGG1 is the main BER enzyme that repairs 8-oxoG, which is a critical mutagenic lesion. The level of OGG1 expression may be modulated by a variety of stimuli including oxidative stress [46]. The transcription factor NRF2 has emerged as a master regulator of intracellular redox homeostasis by controlling the expression of a battery of redox-balancing antioxidants and phase II detoxification enzymes. NRF2 activation has been shown to mitigate a number of pathologic mechanisms associated with Alzheimer's disease, Parkinson's disease, amyotrophic lateral sclerosis, Huntington's disease, and multiple sclerosis [47]. It was reported that some antioxidants (Vit C and BHA) mediated the induction of NRF2 that is involved in the regulation of OGG1 [47]. The human OGG1 promoter contains a putative NRF2-binding site and NRF2 leads to OGG1 transcriptional activation [48, 49]. In this study, we found that pesticide treatment and in a more extent, polyphenols of honey extracts induced nuclear translocation of NFR2 and a significant increase of OGG1 protein expression into the nucleus (cf. Fig. 5). Although either oxidants or antioxidants induced NFR2-mediated nuclear OGG1 expression, increased DNA repair activity was found only in cells treated with honey extracts. These findings suggest a regulatory mechanisms at the posttranscriptional level as previously described [50]. We can postulate that pesticides may affect DNA repair enzyme activity by altering the redox-sensitive site(s) of proteins, which may be prevented by polyphenols of honey extract incubation.

The effect of honey on pesticide-induced genotoxicity was then confirmed in vivo. An altered DDR was found in a residential population long-term exposed to low-level pesticides. In these subjects, a reduced DNA repair activity was observed, with respect to subjects living in a pesticide-free area. Notably, the low DNA repair activity was further reduced in the period of exposure to pesticides, which was associated with a dose-dependent increase of DNA lesions (FPG-sites, ENDOIII-sites). The two-week honey supplementation significantly reversed DNA repair activity and inhibited pesticide-induced DNA damage (cf. Fig. 6). There is an increasing concern about chronic low-level pesticide exposure; a metaanalysis indicated that children exposed to indoor insecticides would have a higher risk of childhood hematopoietic cancers [51]. Prolonged, low-level pesticide exposure can also cause peripheral neurotoxicity, especially in sensory nerves [52]. Thus, highlighting the importance of a diet-rich in polyphenols in the prevention of the deleterious effect of chronic exposure to pesticides.

Taken together, pesticides, even at low doses, induce DNA damage through mitochondria destabilization. DNA lesion accumulation occurs as consequence of a reduced DNA repair activity induced by pesticide exposure. The honey containing polyphenols affects in vitro and in vivo the DNA damage response, thus limiting the toxicity induced by pesticides.

R.A., M.T., N.M. study design and subject enrolment; N.M., S.G., V.C., F.M., performed experiments, data collection, and analysis; M.F.C., F.P. performed the polyphenol profile evaluation on honey extracts; B.B., M.B., M.A. blood and air sample collection, environmental analysis, and contribute for materials and reagents.

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